

AFL Match & Player Prediction — Data Dictionary & Target Definitions

1. Source Data Grain

Dataset	Row Represents	Grain	Primary Entity	Join Key
afl_players_info_raw	One AFL player	Player-level	Player	player_id
afl_players_round_by_round_stats_raw	One player performance record for a match/round	Player-match/round-level	Player × Match	player_id + match_key + season/round
afl_players_seasonal_stats_raw	One player's stats aggregated for a season	Player-season-level	Player × Season	player_id + season
team_matches_home_away_raw	One team's perspective of a match	Team-match-level	Team × Match	match_key + team

2. Feature Dictionary — task4_features_v1.csv

All 12 features below are leakage-safe: each is computed only from information available strictly before the match being predicted. Identifier columns carried alongside these features: match_key, match_date, team_name, opponent.

Feature	Description	Window	Source Columns
is_home	Whether the team is playing at home for this match	Current match	home_away_clean
win_rate_last_5	Team's win rate over its previous five matches before the current one	Previous 5 matches	result_clean, team_name
avg_score_last_5	Average team score over the previous five matches	Previous 5 matches	team_score, team_name
avg_opponent_score_last_5	Average score conceded by the team over its previous five matches	Previous 5 matches	opponent_score, team_name
avg_margin_last_5	Average scoring margin over the previous five matches	Previous 5 matches	margin, team_name
current_win_streak	Number of consecutive wins entering the current match	All previous matches	result_clean, team_name
h2h_win_rate	Team's historical win rate against this specific opponent before the current match	Previous head-to-head matches	team_name, opponent, result_clean

days_rest	Number of days since the team's previous match	Previous match	match_date, team_name
previous_ladder_points	Team's cumulative ladder points before the current match	Pre-match season-to-date	ladder_points, season, team_name
previous_ladder_position	Team's ladder rank before the current match	Pre-match season-to-date	previous_ladder_points, season, round
team_avg_player_disposals_last_5	Average player disposals aggregated to team level from the previous five games	Previous 5 player-games	disposals, player_id, team
team_avg_player_goals_last_5	Average player goals aggregated to team level from the previous five games	Previous 5 player-games	goals, player_id, team

3. Target Definitions

Target	Definition	Aggregation Level	Problem Type
match_outcome	Match result from the selected team's perspective (Win / Loss / Draw), mapped to {win: 1, draw: 0, loss: -1} as the modeling label.	Match (team-perspective)	3-class classification (as implemented)
top_disposal	1 if the player recorded the highest disposals in the match, else 0.	Player-match	Binary classification
top_goal_kicker	1 if the player recorded the highest goals in a scoring match (max goals > 0), else 0.	Player-match	Binary classification
top_fantasy_player	1 if the player recorded the highest fantasy points in the match, else 0.	Player-match	Binary classification

Known limitation: the modeling-decision markdown frames match_outcome as a match-level 3-class target (Home Win / Away Win / Draw). The implemented target is currently the team-perspective Win/Loss/Draw label (match_outcome above); the match-level Home/Away directional version (via determine_match_winner) is exploratory and not yet wired into the feature/target pipeline. Resolve before Day 2 modeling begins.

Leakage rule: same-match player statistics (disposals, goals, fantasy_points) must never be used as features for the corresponding player-match target.